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ORIGINAL ARTICLE

Resolving Perplexity: Comparison of Endoscopic Dacryocystorhinostomy With and Without Stent

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Abstract Endoscopic dacryocystorhinostomy (DCR) is an effective surgical procedure to treat nasolacrimal duct obstruction. This study was conducted with an aim of comparing the success rate between use of stent and without use of stent in endoscopic DCR. A prospective randomized study was conducted. Total 50 cases with signs of nasolacrimal duct blockage were included. The cases were randomly divided in two groups with 25 cases in each group. Group A cases underwent endoscopic DCR with stent and group B without stent. The follow up was till 12th week. Both subjective and objective outcomes were noted. By 12th week, only 8% cases had no relief of symptoms in group A while 92% cases of group A and all cases of group B had complete relief of symptoms. In objective outcome, by the 12th week, in group A 92% cases had full patency while in group B 100% cases had full patency. Overall complications in postoperative period were seen in 28% patients in group A and 12% patients in group B. In this study the surgical results of endoscopic DCR with or without stent came almost equal with no statistical difference in the success rate between stent group and non-stent group. Now that Endoscopic DCR without stent is equally effective and reduces cost, we recommend that the endoscopic DCR without stent should be preferred.

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Keywords Nasolacrimal duct · Endoscopic dacryocystorhinostomy · Stent

Introduction

Epiphora is an imperfect drainage of lacrimal secretions through the lacrimal passage. The most common cause being chronic dacryocystitis most commonly due to nasolacrimal duct obstruction [1]. External dacryocystorhinostomy (DCR) was gold standard treatment for this till few decades back after which endoscopic approach came and took over the conventional method. An Italian ophthalmologist Toti described the first external approach to dacryocystorhinostomy in 1904 [2]. This procedure was modified with the use of flaps by many authors and Dupuy Dutemps and Bourguet described the modern external flap DCR technique [3]. The main disadvantage with all these techniques was an external scar. Caldwell first described the intranasal DCR in 1893 [4], which was later modified [5]. McDonogh and Meiring were the first who used endoscopes in trans nasal dacryocystorhinostomy in 1989 [6]. This recent revolutionary method known as endoscopic DCR for the treatment of chronic dacryocystitis has been proved promising. Endoscopic DCR has been established as the prevailing procedure for acquired nasolacrimal duct obstruction [7, 8]. In some of the previously published studies, it has been claimed that stent improves the surgical outcomes [9–11]. On the other hand, some studies indicate that stent itself is the reason for surgical failure [12, 13]. From the time of advent of the surgery, the use of stent is still controversial. So, this prospective randomized study has been undertaken to compare the success rate of endoscopic DCR with and without stent.

Materials and Methods

This prospective study was conducted in the department of ENT, Head and Neck surgery at Pramukhswami Medical College and Shree Krishna Hospital, Karamsad, Anand and ENT hospital, Doctor house, Anand during the period of June 2010 to September 2012. The patients of either sex and with different age groups with nasolacrimal duct obstruction diagnosed based on history, clinical examination and sac syringing was included in the study. Patients with blockage at the level of canaliculi, with epiphora due to lacrimal pump failure, with bony deformity of lacrimal fossa and patients less than 3 years of age were excluded from the study. Total forty-eight patients were included. Two patients were operated bilaterally so each eye which was operated has been considered as one "case" making it to total 50 cases being included in the study. The cases were randomly divided in two groups (group A and group B) with 25 cases in each group. The group A cases underwent endoscopic DCR with stent and group B cases endoscopic DCR without stent after obtaining written informed consent.

Initial patient work-up included detailed history about the symptoms. Thereafter, detailed examination including anterior rhinoscopy, throat and ear examination with local eye examination was done. Nasolacrimal duct and lacrimal sac obstruction was confirmed by sac syringing. All patients who demonstrated regurgitation through opposite punctum were taken up for the surgery after the routine investigations. A Performa was prepared for recording the findings. Surgery was performed under general anesthesia using 0° and 30° endoscopes. The lacrimal silicone stent was put in group A patients (Fig. 1). All patients were discharged on regime of oral antibiotics and anti-inflammatory drugs, nasal decongestant and local antibiotic eye drops. Regular follow-up of patients was done at 1, 3, 6 and 12 weeks. The stent was removed at 6th postoperative week. Subjective evaluation was done based on symptomatic relief of the symptom of watering of eyes in terms of complete relief, partial relief and no relief. Objective evaluation was done by sac syringing and results were evaluated as patent, partially patent and blocked nasolacrimal duct. Considered as patent if fluid passed into the nasopharynx without reflux out of the opposite punctum, partially patent if some of the fluid regurgitated through the opposite punctum and some passed into the nasopharynx and blocked if no fluid passed into nasopharynx but whole fluid regurgitated through either punctum. The statistical significance (p value) was calculated using the Chi square test.

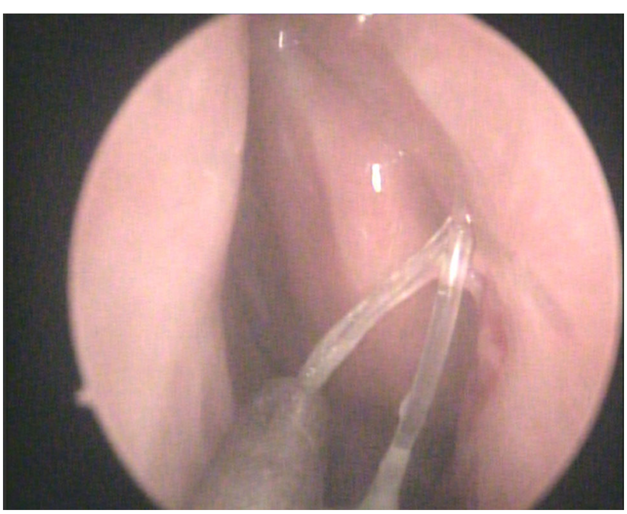

Fig. 1 Shows the presence of Lacrimal stent on endoscopic examination

Results

The findings in 50 cases of chronic dacryocystitis who underwent endoscopic DCR with and without stent were analyzed. The cases were randomly divided into two groups with 25 cases in each group (group A with stent and group B without stent). It was observed that the mean age in the study was 50.04 years. Overall 65.5% females and 35.4% males were involved. Left eye was involved in 52% cases and right eye in 48% cases. In group A 23 cases (92%) were diagnosed as chronic dacryocystitis while 2 cases (8%) were of chronic dacryocystitis with mucocele. In group B, there were 24 cases (96%) of chronic dacryocystitis and one case (4%) was of chronic dacryocystitis with mucocele. The commonest presenting symptom was epiphora which was present in all patients. Other symptoms noted were discharge from the eye in 8% cases in group A and 16% in group B while swelling over the lacrimal sac area in 16% in group A and 20% in group B. All patients were regularly followed at 1, 3, 6, 12 weeks. Subjective evaluation was made in terms of complete, partial or no relief from symptoms. Complaint of minimal watering was present in 6 cases (24%) in stent group (group A) and 5 cases (20%) in non-stent group (group B) by 1st week and 4 cases (16%) in group A with 3 (12%) in group B by 3rd week. By the 6th week only 2 cases (8%) had no relief of symptoms in group A while rest of the 23 cases (92%) of group A and all 25 cases (100%) of group B had complete relief of symptoms which was continued to be the same till the end of 12th week. Considering the short term results the functional success was 68% in the stent group and 80% in non-stent group at 1st week follow up (p value 0.311) (Table 1). The success rate increased to 92% in stent group and 100% in non-stent group by 12 weeks follow up (p value 0.149) (Table 2).

Table 1 Subjective evaluation at 1st week ($p = 0.311$)

Watering from eyes	With stent (group A)	Without stent (group B)
Complete relief	17	20
Partial relief	6	5
No relief	2	0

Table 2 Subjective Evaluation at 12th week ($p = 0.149$)

Watering from eyes	With stent (group A)	Without stent (group B)
Complete relief	23	25
Partial relief	0	0
No relief	2	0

The findings of sac syringing on followup showed full patency in 76% cases in group A and 88% cases in group B by 1st week and 80% in group A with 92% in group B by the end of 3rd week. On further follow up in stent group (group A) 2 cases (8%) had complete blockage, rest all 23 cases (92%) had full patency on sac syringing while in non-stent group (group B) all patients had full patency by the end of 6th week and towards the completion of 12 weeks. So, on objective evaluation the success rate was 76% in stent group and 88% in non-stent group by the 1st week follow up (p value 0.307) (Table 3). The success rate increased to 92% in stent group and 100% in non-stent group by 12 weeks (p value 0.149) (Table 4).

No major complication occurred in our study. In stent group four patients had granulations and obstruction over the rhinostomy site, three patients had synechiae formation. In without stent group only one patient had mild granulation at the stoma site and two patients had synechiae formation in the nasal cavity. Overall complications in postoperative period were seen in 28% cases in stent group as compared to 12% cases in without stent group ($p = 0.490$) (Fig. 2).

Table 3 Objective evaluation at 1st week ($p = 0.307$)

Sac syringing	With stent (group A)	Without stent (group B)
Fully patent	19	22
Partly patent	4	3
Blocked	2	0

Table 4 Objective evaluation at 12th week ($p = 0.149$)

Sac syringing	With stent (group A)	Without stent (group B)
Fully patent	23	25
Partly patent	0	0
Blocked	2	0

Discussion

Endoscopic Dacryocystorhinostomy has been gaining popularity recently in last one decade compared to conventional external dacryocystorhinostomy. Some recent manuscripts have shown that Endoscopic DCR success rates are equal to or better than external DCR. In 2007 Yigit et al. [14], compared external and endoscopic DCR on a total of 103 patients and concluded that Endoscopic DCR is safe and reliable procedure.

Since its advent, endonasal endoscopic DCR has evolved like anything. Various newer techniques have been evolved in endoscopic DCR. The use of powered instruments have given excellent results as shown in a study done in 2007 by Ramkrishnan et al. [15] as they showed excellent outcomes of powered endoscopic dacryocystorhinostomy without the preservation of mucosal flaps for the management of acquired nasolacrimal duct obstruction. The latest advancement is the successful use of laser in endoscopic DCR. The use of laser to create stoma was first described by Massaro et al. in 1990 using argon laser [16]. The holmium-YAG laser was first used by Metson et al. [17] in a series of 40 patients. With the evolution of techniques the laser surgery has been made even more précised with the use of miniendoscopes which enabels endoscopy of the lacrimal drainage system via the lacrimal puncta to visualize the exact site of a stenosis. Recently in 2003 a study by Thiemo Hofmann et al., on a consecutive sample of 78 patients who underwent potassium-titanyl-phosphate (KTP) LASER assisted dacrycystorhinostomy with the use of modified lacrimal miniendoscope concluded that the success rate of 83% achieved with KTP LASER assisted dacryocystorhinostomy using miniendoscopes for lacrimal surgery to visualize the exact site of obstruction, which is better compared with that of prior studies without the use of miniendoscopes [18].

Although Endoscopic DCR has revolutionized the surgery for nasolacrimal duct blockage with its many distinct advantages like it prevents external scar, shorter surgical time, acute stage is not a contraindication, causes comparatively less morbidity and any additional procedure required can be done at the same sitting [12, 19], stenosis or closure of the opening is one of the common causes of failure. Although there are number of studies on the use of stents to prevent stenosis, there is no consensus regarding their efficacy.

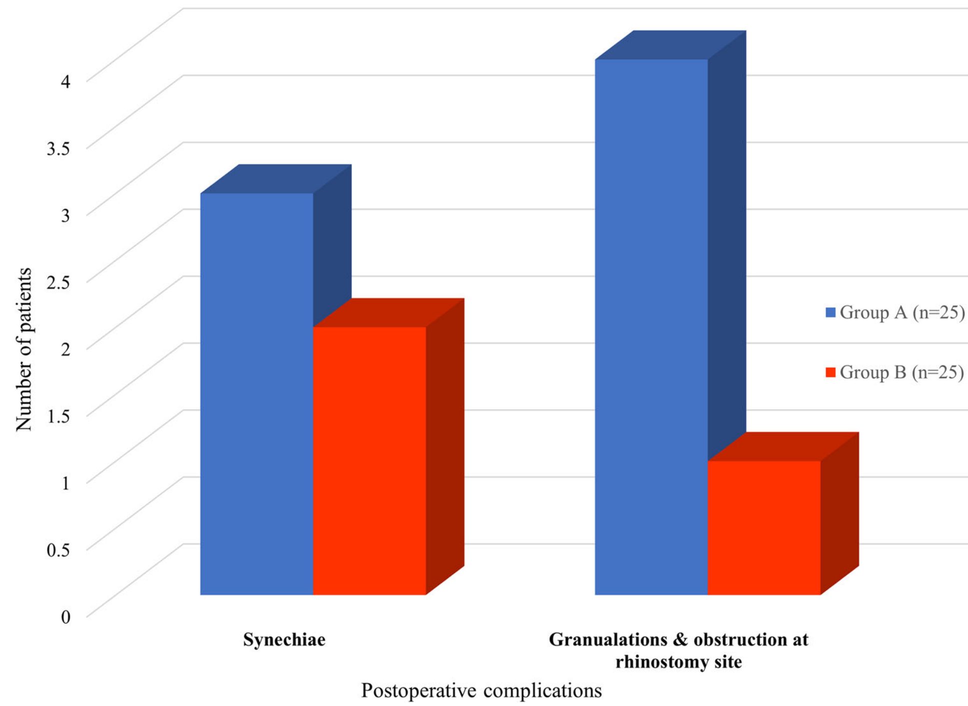

Fig. 2 Distribution of postoperative complications in cases (N = 50) ($p = 0.490$)

This study was undertaken to compare the results of endoscopic DCR with and without stent. In the present study, the mean age of 50.4 years and female preponderance was seen which is same as seen in previous studies [14, 20]. In the present study, out of total 50 cases, 26 (51%) were operated for right side nasolacrimal duct obstruction and 24 (49%) were operated for left side. It is almost like that observed by Yigit et al. [14] and Pittore et al. [20] but as such this disease has no special predilection to the laterality. In subjective evaluation, by the 3rd week three cases in without stent group and four cases in stent group had only partial relief of symptoms. In without stent group, in two cases it was found to be due to crusting in the nasal cavity which was removed endoscopically and regular follow up with sac syringing achieved complete relief while in one patient it was mainly due to mild granulations at the rhinostomy site. It was found to be minimal and was removed and complete relief could be achieved. In stent group, two patients were having minimal granulations which could be taken care of and complete relief achieved by the 6th week while other two patients with granulations and stenosis over rhinostomy site there was no relief at all and had to undergo revision surgery. Thus, except two cases of stent group all other had complete relief of symptoms by the end of 6th week which was continued till 12th week (Table 3). The analysis of objective evaluation showed that in group A (stent group) two cases who were having block even at the 6th week, by the end of 12 weeks 23 cases (92%) were patent on sac syringing (Table 3). In group B (without stent group) at the end of 3 months all 25 cases (100%) were fully patent on sac syringing (Table 4). This was also evident on postoperative nasal endoscopy at 12 weeks follow up, evaluating anatomical patency (Fig. 3).

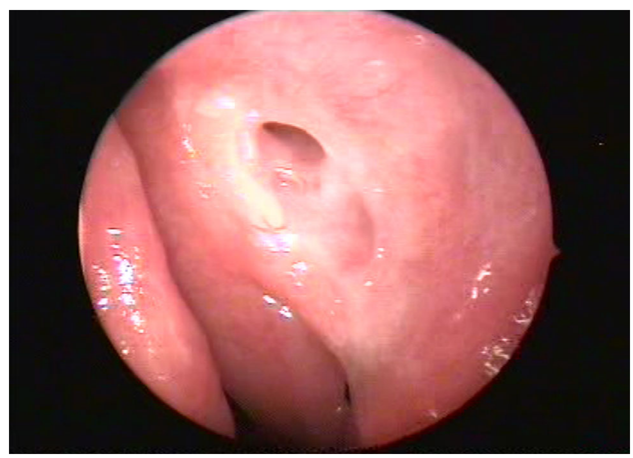

Fig. 3 Endoscopic examination of rhinostomy site at 12 weeks follow up

In two patients of stent group (out of total four) and one patient of without stent group it was seen that the granulations or synechiae around the stoma were minimal. These granulations were removed by local cauterization and had otherwise not yet caused the blockage of stoma. As seen in two patients of with stent group minimal granulations could be taken care of and good patency of the lacrimal pathway could be achieved. But the other two patients in stent group were having blockage even at the end of

12 weeks and had to undergo revision surgery. They were having severe obstruction at the rhinostomy site because of granulations or synechiae, which were probably due to the use of the stent and can be attributed to foreign body granuloma. The cause of granulation or synechiae in one patient of without stent group might be excessive removal of the mucosal flap and thus more bone exposure which got healed with time and the patient got relieved of the symptoms. But it was observed that adequate removal of the mucosal flap with high drilling of the thick bone gives good exposure of the lacrimal sac and thus gives good results.

Overall 28% cases showed complications in stent group as compared to 12% patients in without stent group which came out as statistically insignificant. It is in correlation with other studies in literature as Nayak et al. [19] had three cases of synechiae formation and two had granulations in the operated area which was successfully treated endoscopically as an office procedure. Even Jin et al. [21], reported in their study that in 17% cases rhinostomy opening was found to be obstructed by granulations or synechiae with primary success rate of 83% with endoscopic DCR with stent. There were different other complications reported in other studies like canaliculi erosion due to overly tight stent, lacrimal pump failure, corneal abrasion, orbital injury, and CSF leakage through fractured ethmoid [22, 23] which were not seen in the present study. Although controversial, stents are used to keep the neo-ostium open but it has been seen that it adds to complications [23] and adds the risk of granulation formation and infection at the site [24].

Allen and Berlin reported in their study that silicone intubation at the time of DCR was associated with a statistically significant increase in the failure rate of primary DCR [25]. While in our study, there was 92% success rate in cases who underwent endoscopic DCR with stent and 100% in cases who were taken for endoscopic DCR without stent. Although no statistical difference (p value 0.149) between the two groups, the success rate in cases without stent is comparable with high success rates in other studies as Singh et al. [26] also reported success rate of 92.6% of endoscopic DCR without stent with no major complications. This has been reported in another randomized trial by Unlu et al. [27] showing results in the stented and non-stented groups at 84.2 and 94.7%. Although there was no statistical difference showing one technique to be superior to another, the authors recommended that stenting was not a mandatory requirement following endoscopic DCR.

Smirnov demonstrated in a prospective randomized trial of patients undergoing endoscopic DCR that silicone stenting is unnecessary, giving an overall success rate with and without stent of 78 and 100% respectively [28]. Harvinder et al. [29] showed powered endoscopic DCR with mucosal flaps without stenting has a success rate comparable to that achieved with stents. In a very latest study of 2016, Longari et al. [30] have shown Success rate after 18 months follow-up were, respectively, 82.2% in the stent group and 88.6% in the non-stent group with no statistical differences and concluding that Endoscopic DCR without silicone stent intubation should be the first choice of procedure, stent intubation should be reserved in selected cases with poor local conditions pre and intra-operatively assessed. Although Endoscopic DCR with and without stent are showing good success rate the success of all procedures is not mandatory, particularly the revision surgeries [31, 32]. The most common reason for the failure is the formation of granulation tissue or scarring over the rhinostomy site. It has been shown that the intraoperative topical use of antiproliferative mitomycin C may reduce postoperative scarring and improve the success rate in revision Endoscopic DCR [33].

Conclusion

In this study the surgical results of endoscopic DCR with or without stent came almost equal with no statistical difference in the success rate between stent group and non-stent group. Now that Endoscopic DCR without stent is equally effective and reduces cost, we recommend that the endoscopic DCR without stent should be preferred.

Compliance with Ethical Standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

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